

Intestinal Barrier in Post-Campylobacter jejuni Irritable Bowel Syndrome.

PMID: 36979384 | Indexed in PubMed

Campylobacter jejuni (*C. jejuni*) is one of the most common causes of bacterial gastroenteritis worldwide. One sequela of this infection is the development of post-infectious irritable bowel syndrome (PI-IBS). It has been suggested that a dysfunctional intestinal barrier may promote IBS development. We aimed to test this hypothesis against the background of the leaky gut concept for low-grade inflammation in PI-IBS. We identified patients with persistent PI-IBS symptoms after *C. jejuni* infection. During sigmoidoscopy, forceps biopsies were obtained for electrophysiological measurements of epithelial transport and barrier function in miniaturized Ussing devices. *C. jejuni* absence was checked by PCR and cytokine production with immunohistochemistry. In PI-IBS, the epithelial resistance of the colon epithelium was unaltered, reflecting an intact paracellular pathway. In contrast, temperature-dependent horseradish peroxidase (HRP, 44 kDa) permeation increased. Short-circuit current (*I*_{sc}) reflecting active anion secretion and ENaC-dependent electrogenic sodium absorption was unaffected. Early endosome antigen-1 (EEA1) and IL-4 levels increased. *C. jejuni* is not incorporated into the resident microbiota of the colon mucosa in PI-IBS. In PI-IBS after *C. jejuni* infection, macromolecule uptake via endocytosis was enhanced, leading to low-grade inflammation with pro-inflammatory cytokine release. The findings will allow *C. jejuni*-induced pathomechanisms to be targeted during infection and, thereafter to reduce sequelae such as PI-IBS.

Blastocystis subtypes and their association with Irritable Bowel Syndrome.

PMID: 29857906 | Indexed in PubMed

Blastocystis spp. is a common intestinal protozoan that affects humans and animals. The role of this parasite as a pathogen is still controversial and it is suspected to be linked to Irritable Bowel Syndrome (IBS), a functional bowel disorder characterized by chronic or recurrent abdominal pain associated with altered intestinal habits. A broad search in electronic databases, libraries, portals of journals, etc. for reports on the association between the parasite and IBS without language restriction was performed. The selection was not restricted by date, but articles published in the last seven years were given preference. We investigated the evidence regarding *Blastocystis* and IBS coexistence as well as the implications of the parasite in pathogenesis and clinical manifestations. Only standardized parasitological tools, supplemented by epidemiological analysis, will be able to clarify whether parasite carriage could be connected to IBS and its status as a human pathogen. Although a variation in pathogenicity and virulence between subtypes has been confirmed, *Blastocystis* can only be considered an indicator of dysbiosis. Accurate diagnoses of this parasitic eukaryote, specifically at genotypic and phenotypic levels, as well as the complete analysis of the intestinal microbial communities, have to be included in the protocol of those patients with IBS.

Effectiveness of sublingual immunotherapy in pediatric cedar pollinosis: A real-world database study.

PMID: 38284920 | Indexed in PubMed

Pediatric allergic rhinitis (AR), including cedar pollinosis (CP), is increasing in Japan. We investigated the effects of sublingual immunotherapy (SLIT), which has limited studies of its effectiveness in real-world

settings, on children with CP. This retrospective cohort study used a claim database in 2018-2021. Children aged ≤ 15 years with CP records in 2019 were eligible and were followed up through 2021. We included 2962 CP children undergoing SLIT and 547 who were not. The medication score was used to evaluate SLIT effectiveness in the cedar pollen dispersal season each year. Adverse events and the occurrence of allergic diseases were also evaluated. Medication score was higher in the SLIT group during the index period but lower in 2021 compared to the non-SLIT group (mean $\pm$ standard deviation: 5.17 ± 2.39 and 4.74 ± 2.38 in 2019, 3.13 ± 2.30 and 3.55 ± 2.48 in 2021, respectively). The adjusted mean difference between groups from 2019 to 2021 was -0.62 (95% confidence interval: -0.86 to -0.39 , $p < .0001$), and the medication score was reduced in the SLIT group (risk ratio: 1.2: 1.1 to 1.3). The occurrence of adverse events involving abdominal disorders (adjusted odds ratio [aOR]: 0.64: 0.51 to 0.81), asthma exacerbation (aOR: 0.37: 0.24 to 0.57), and allergic diseases involving hay fever unrelated to CP (aOR: 0.60: 0.45 to 0.80) or asthma (aOR: 0.71: 0.58 to 0.86) was lower in the SLIT group. In children with CP, SLIT is effective, well tolerated, and could decrease the occurrence of other allergic diseases.

Sublingual immunotherapy for cedar pollinosis possibly triggers eosinophilic esophagitis.

PMID: 38032451 | Indexed in PubMed

Sublingual immunotherapy (SLIT) is an effective and popular treatment for cedar pollinosis. Although SLIT can cause allergic side effects, eosinophilic esophagitis (EoE) is a lesser-known side effect of SLIT. A 26-year-old male with cedar pollinosis, wheat-dependent exercise-induced anaphylaxis, and food allergies to bananas and avocados presented with persistent throat itching, difficulty swallowing, heartburn, and anterior chest pain 8 days after starting SLIT for cedar pollinosis. Laboratory examination showed remarkably elevated eosinophils, and esophagogastroduodenoscopy revealed linear furrows in the entire esophagus. Histological examination of an esophageal biopsy specimen revealed high eosinophil levels. The patient was strongly suspected with EoE triggered by SLIT. The patient was advised to switch from the swallow to the spit method for SLIT, and the symptoms associated with SLIT-triggered EoE were reduced after switching to the spit method. This case highlights the importance of recognizing SLIT-triggered EoE as a potential side effect of SLIT for cedar pollinosis, especially with the increasing use of SLIT in clinical practice. EoE can occur within a month after initiating SLIT in patients with multiple allergic conditions, as observed in our case. Furthermore, the spit method should be recommended for patients who experience SLIT-triggered EoE before discontinuing SLIT.

Phenological Analysis of Grasses (Poaceae) in Comparison with Aerobiological Data in Moscow (Russia).

PMID: 39273868 | Indexed in PubMed

Grasses (Poaceae) produce large amounts of pollen and are among the main causes of pollinosis worldwide. Despite their morphological similarity, pollen grains of different grass species may have different allergenicities. Therefore, quantification of the roles of individual species in airborne pollen is an important task. There are very few studies on this topic, and none of them have been conducted in a temperate continental climate. Our study was carried out for three years (2020-2022) in the urban

territory of Moscow (Russia) and aimed to understand what grass species contribute the most to the total pollen load of the atmosphere. The comparison of aerobiological and phenological data was based on calculating the phenological index, which is a combination of phenological parameters, pollen productivity of individual species, and their abundance. Our data showed that the decomposition of pollination curves based on the phenological index was sometimes very efficient but not always possible in temperate continental climates. The main reasons for disagreement between aerobiological and phenological data were weather conditions and lawn mowing. Not all grasses were equally important as sources of allergenic pollen. The greatest contribution to the pollen load at the beginning of the season in Moscow was made by *Dactylis glomerata*, and to a lesser extent by *Phleum pratense* and *Festuca pratensis*. These are the most common species, which are widespread throughout Europe. The contribution of minor components is insignificant.

Cross-reactivity to olive tree pollen and orchard grass pollen in patients with pollinosis.

PMID: 9227797 | Indexed in PubMed

We studied 92 patients with allergic rhinitis in Syodosima, Japan, during the pollen season between April and June to evaluate the cross-reactivity to different antigens, including pollen from the olive tree (*Olea europaea*) and orchard grass (*Dactylis glomerata*). Olive tree pollen was present in the atmosphere for 23 days, from May 19 to June 12, 1994. Specific IgE antibodies for olive tree pollen antigen were present in 21 (26.9%) of the 78 patients with allergic rhinitis. Nine (24.3%) of the 37 patients with allergic rhinitis exhibited positive skin reactivity to an extract of olive tree pollen. Fifteen (88.2%) of the 17 patients who had IgE reactivity in their sera to olive tree pollen antigen demonstrated allergic reactions to an extract of olive tree pollen. Specific IgE antibodies for orchard grass pollen antigen were present in 43 (48.3%) of the 89 patients with allergic rhinitis and 20 (95.2%) of the 21 patients who had IgE reactivity in their sera to olive tree pollen antigen. The inhibition test using the CAP System revealed that the reactivity of the IgE antibody specific for olive tree pollen antigen was inhibited dose-dependently by an extract of orchard grass pollen. These findings show that there is a reaction in some patients with grass (Gramineae) pollinosis that might be induced by olive tree pollen.

Pollen allergic disease: pollens and its major allergens.

PMID: 17143438 | Indexed in PubMed

Patients with grass pollen allergy, commonly called pollinosis, often present reactivity to pollen allergens from a number of grass species due to cross-reactivity of IgE antibodies to pollen proteins present in pollen grasses. In this context, Italian rye grass (*Lolium multiflorum*) pollen of the Poaceae family cultivated in Southern Brazil has been considered a major sensitizing agent in patients with pollinosis. In this region, Italian rye grass is capable of producing a great amount of pollen. In addition to *L. multiflorum*, other Poaceae grasses are naturally grown as weed in Southern Brazil, but with no clinical relevance. Pollen extracts derived from homologous or heterologous grasses are often used for both diagnosis and treatment of seasonal allergy. However, no standardized *L. multiflorum* pollen extract is commercially available in Brazil and mixed grass extracts are commonly used for diagnosis and immunotherapy of grass pollen allergy. Further studies are required to better characterize the cross-reactivity between *L. multiflorum* and other grass pollen allergens for improving the diagnosis and immunotherapy to *L. multiflorum* pollen allergy.

[Pollen as the cause of allergies].

PMID: 11407229 | Indexed in PubMed

Pollinosis or hay fever is the most common allergic disease in Switzerland. For symptoms during spring pollens of birch and related trees (alder, hazel) and also ash tree are responsible, while hay fever during summer is mainly caused by pollens of grasses, rye and mugwort. These main plant pollen allergens, relevant cross-reactivities with other pollen and food allergens are reviewed in this article. The well-established methods of pollen-counting in Switzerland allow to define the varying amounts of measurable pollen depending on geographic and climatic conditions. Similarly clinical symptoms, the diagnostic work-up of pollen allergies and therapeutic aspects including preventive measures, symptomatic therapy and specific immunotherapy are presented. Finally, occupational and travelling aspects of pollinosis are briefly discussed.

[Effect of oak pollen on patients with birch pollinosis].

PMID: 7738699 | Indexed in PubMed

The author reviewed 53 cases of birch pollinosis treated in our ENT-clinic between 1990 and 1991. Of interest is the fact that many of these patients complained of typical symptoms in and after June, when birch pollen has usually disappeared. We considered this discrepancy to be attributable to the effects of grass pollen, one of the common causal agents in this period. However, only a few patients tested positive for grass pollinosis. We therefore postulated that cross-reactivity between birch and oak pollen accounted for this phenomenon, because both trees belong to same order, Fagales. A strong correlation between RAST scores for birch and oak pollen was detected, and the oak pollen RAST score was significantly reduced by birch pollen extract in an inhibition test. These findings demonstrated actual cross-reactivity between birch and oak pollen. Birch pollinosis patients in Sapporo therefore have allergic symptoms from March to July.

[Influence of sensitization to pollen and food allergens on pollinosis clinical symptoms].

PMID: 15827387 | Indexed in PubMed

Geographic position and local plants of the country influence the profile of sensitization of the population to airborne allergens. The aim of this study was to evaluate the sensitization pattern to pollen and food allergens in adult patients with pollinosis in Lithuania. 101 patients (age 16-63 years) suffering from seasonal allergic rhinitis and 23.8% of them also diagnosed with concomitant seasonal asthma were investigated. Oral allergy syndrome (OAS) was diagnosed in 29.7% of cases. The sensitization to 21 species of tree-, grass- and weed-pollen and plant food allergens was determined by positive skin prick and prick-prick test. In serum levels of total IgE and timothy and orchard grass specific IgE were determined by immunoenzyme assay. 52.5% of patients suffered from spring-summer pollinosis. 91.2% of patients were sensitized to grass-pollen allergens, 79.3% -to tree pollen-allergens. 74.7% of patients were allergic to weeds. Pollinosis starting in the spring and lasting more than sixteen weeks was associated with increased probability of OAS (OR=7.1, $p<0.001$ and OR=3.1, $p=0.01$). Sensitization to hazelnut (OR=8.6, $p=0.009$), birch (OR=9.6, $p=0.07$), lamb's quarters (OR=5.2, $p=0.04$) allergens and

twofold and more increase in serum IgE ($OR=4.8$, $p=0.03$) were considered the significant risk factors for pollinosis with OAS. More than two times elevated serum IgE increased the probability of seasonal asthma ($OR=3.4$, $p=0.03$). Sensitization to ragweed was associated with decreased risk for asthma ($OR=0.26$, $p=0.03$). Our data indicate that more than a half of patients (52.5%) had pollinosis symptoms during spring and summer seasons because of multiple sensitivity to pollen allergens. Sensitization to hazelnut, birch, lamb's quarters allergens, more than two times elevated serum IgE are significant risk factors for pollinosis with OAS. More than two times elevated serum IgE increased the probability of seasonal asthma, but sensitization to ragweed was associated with decreased risk for pollinosis with asthma.
